

CAN I BELIEVE MY EYES?

We may think we see with our eyes, but we actually see with our brains. When light enters the eye, millions of tiny sensors send nerve signals at high speed along the optic nerve to the brain, which then interprets them to give a three-dimensional (3-D) full-color window on to the world. Most of the time, this works very well, but sometimes the brain is tricked or fooled into seeing something that isn't really there, so watch out!

ON THE LOOKOUT

CLEVER CAMOUFLAGE

Animal camouflage is nature's way of tricking the eye. To fool predators, a stick insect that looks just like a twig or a frog blends in perfectly with the tree bark on which it lives. When zebras graze in a group, their stripes have the effect of merging their bodies together, so a hungry lion would find it difficult to pick out and hunt down an individual animal.

MISLEADING MIRAGES

A thirsty traveler in the desert sees a pool ahead. He hurries toward it—but it vanishes. This is a mirage, caused by a layer of hot air above the sand. Light rays entering the warm air are refracted (bent) before entering the eye. What is hitting his retina (the light-sensitive membrane at the back of the eye) is an image of the sky, but the man is staring at the ground, so his brain tells him that it is water.